

ASSIGNMENT #07

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Fall 2025
Professor: Dr. Michael Caracotsios
Office: EIB 244
Cell Phone: (630) 696-0934
E-mail: mcaracot@uic.edu

Names : Amaan Choughale and Caleb Murg

Due Date: Friday, November 21st my office (03:00 PM)

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Bring a hard copy of your report as a PDF file by the Due Date and Time in class

- Assignments that are not typed will not be accepted.
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- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

CHE 301

Assignment Number 2

November 21, 2025

Caleb Murg
Amaan Choughale

1.

Problem A: Gas Mixture Saturation with Water

- In the natural gas industry, the amount of water contained in gas streams is given in lb of water per MMSCF of dry gas (lb/MMSCF). Consider a gas stream containing 60.847 wt% methane, 14.256 wt% ethane, 8.363 wt% propane, 8.267 wt% iso-butane and 8.267 wt% normal-butane. What is the maximum amount of water in lb/MMSCF that it can hold at pressure $P = 900$ psia and temperature $T = 65^\circ\text{F}$. Assume that the natural gas mixture with and without water follows the Peng–Robinson Equation of State (EOS) with zero BIPs (Binary Interaction Parameters). The thermal properties of pure water are also calculated from the Peng–Robinson EOS.
- Calculate the dew point and bubble point temperature in degrees Fahrenheit of the saturated natural gas stream at $P = 900$ psia.

Note: The maximum amount of water that a gas mixture can hold at a given temperature and pressure is the quantity of water in the gas mixture that if exceeded, then a drop of pure liquid will be formed at the prevailing temperature and pressure.

Problem B: Gibbs Phase Rule and CO₂ Solubility in Water

A mass of air comes into contact with cold water and equilibrium is established at a temperature of 20°C and pressure of 2 bar. Assume that the gas phase follows the Peng–Robinson EOS and the liquid phase is ideal. Use the steam tables to calculate the saturated vapor pressure of water. The molar composition of dry air can be taken as: 78.084% N₂, 20.946% O₂, 0.928% Ar, and 0.042% CO₂.

- Calculate the Degrees of Freedom by applying the Gibbs Phase Rule and any other constraints present in the system.
- Calculate the solubility of CO₂ in the water phase. Your answer should be given in mg of CO₂ per liter of pure water.

The Binary Interaction Parameters (k_{ij}) for the components in the vapor phase that follow the Peng–Robinson EOS can all be set equal to zero.

Problem C: Mixed Refrigerant Expansion in a Liquid Turbine

Saturated liquid mixed refrigerant (MR) at $P_1 = 60$ bar with composition shown in the table below is expanded down to a pressure of $P_2 = 5$ bar in a liquid turbine. The efficiency of the liquid turbine is 100%.

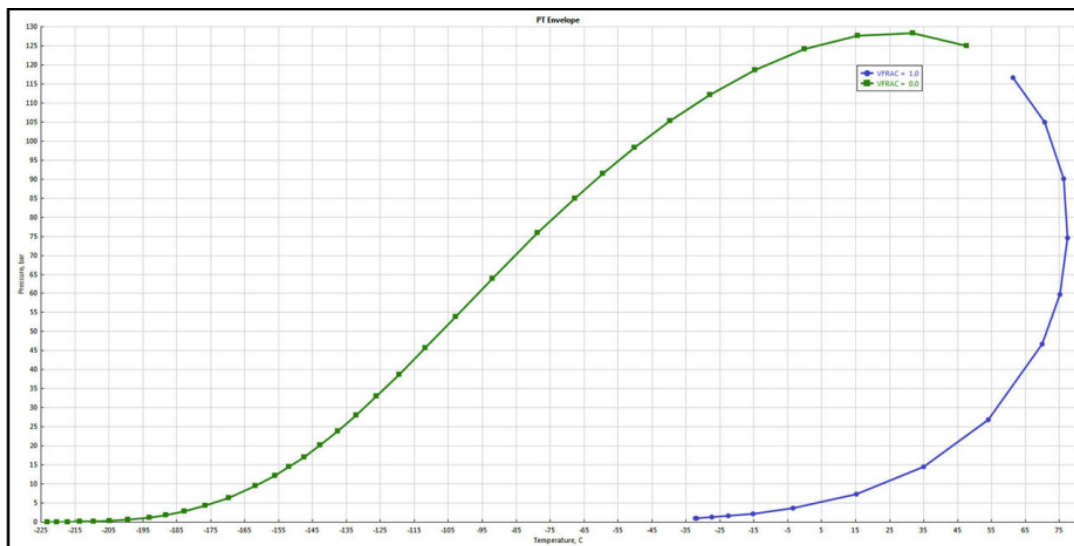
Component	Refrigerant Mole fractions
Nitrogen	0.15
Methane	0.30
Ethane	0.30
Propane	0.00
n-Butane	0.25

You may assume that the refrigerant's thermal properties are described by the Peng–Robinson Equation of State. Binary interaction parameters for the components are given below.

	Component i	Component j	Source	Temperature units	K _{AJ}
▶	METHANE	ETHANE	APV110 EOS-LIT	C	-0.0026
▶	METHANE	PROPANE	APV110 EOS-LIT	C	0.014
▶	ETHANE	PROPANE	APV110 EOS-LIT	C	0.0011
▶	METHANE	BUTANE	APV110 EOS-LIT	C	0.0133
▶	ETHANE	BUTANE	APV110 EOS-LIT	C	0.0096
▶	PROPANE	BUTANE	APV110 EOS-LIT	C	0.0033
▶	METHANE	NITROGEN	APV110 EOS-LIT	C	0.0311
▶	ETHANE	NITROGEN	APV110 EOS-LIT	C	0.0515
▶	PROPANE	NITROGEN	APV110 EOS-LIT	C	0.0852
▶	BUTANE	NITROGEN	APV110 EOS-LIT	C	0.08
+					

Calculate:

1. The temperature (in °C) of the mixed refrigerant at the inlet and outlet of the turbine.
2. The vapor fraction of the mixed refrigerant at the outlet of the turbine.
3. The power generated by the turbine in kJ per kg of mixed refrigerant.



PT Phase Envelope for the Mixed Refrigerant

Ans: A. In the first part of problem A, we were asked to calculate the amount of water in a dry gas stream in lbs per MMSCF (million metric standard cubic feet). To attempt this, we use the fundamental equality of vapor-liquid equilibrium, which is:

$$f_i^v = \bar{f}_i^v$$

Knowing that the fugacity of the pure component in the vapor phase is equal to the fugacity of water in the mixture in the vapor phase, we can set up a balance to solve for the mole fraction of water in the vapor. To do this, we must first recognize that our composition of the dry gas was given in weight percent.

We were told that the gas stream contained 60.847 wt% Methane, 14.256 wt% Ethane, 8.363 wt% Propane, 8.267 wt% Iso-Butane, and 8.267 wt% Normal-Butane. To convert this from weight percent to mole percent, we first assume a basis of 100 kg of dry gas. This allows us to convert from weight percents to weights, where we see we have 60.847 kg Methane, 14.256 kg Ethane, 8.363 kg Propane, 8.267 kg Iso-Butane, and 8.267 kg Normal-Butane. We can now convert to moles of each substance using its molar weight. Our numbers in moles are 3.7927 moles of Methane, 0.4741 moles of Ethane, 0.1897 moles of Propane, 0.1422 moles of Iso-Butane, and 0.1422 moles of Normal Butane. Our total moles can be found by summing the above, and we find that:

$$n_{\text{mol, tot}} = 4.7409 \text{ mols}$$

Using this, we can find the mol% of each compound using the formula

$$\text{mol\%} = \frac{n_i}{n_{\text{tot}}} \cdot 100\%$$

Using the above formula, we find our mol% to be 80.00 mol% for Methane, 10.00 mol% for Ethane, 4.00 mol% for Propane, 3.00 mol% for Iso-Butane, and 3.00 mol% for Normal Butane.

Now that we have the molar composition of the dry gas, we can move on to finding the molar composition of wet gas, where wet gas refers to the fact that the gas is saturated with water. To do this, we need to set an initial guess for the molar composition of water in the stream. We chose 0.001 for our initial guess. Using this initial guess requires us to modify the molar composition of the other components in the stream. This is necessary since including an initial guess for the water will result in a molar composition above 1. To modify the molar compositions of the other components, we use the equation:

$$z_i^o = z_i \cdot (1 - z_{H_2O}^o)$$

where z_i^o is the molar composition of another component in the stream in relation to the guessed value of water composition, z_i is the previously calculated molar composition of another component, and $z_{H_2O}^o$ is our guess for water. In this case, i is one of our other species in our gas stream (Methane, Ethane, Propane, Iso-Butane, and Normal-Butane) and cannot be water. Doing this keeps our total molar composition of the mixture equivalent to 1 as we change the molar composition of water. To find the actual molar composition of water, we need to first find the fugacity of pure water in the vapor phase. This is done by first finding the compressibility of pure water at $P = 900 \text{ psia} = 62.0526 \text{ bar}$ and $T = 65^\circ F = 18.33^\circ C$. We use the `zcomp` function within the CHE Calculator and find the value of the compressibility to be:

$$Z_{H_2O,pure} = 0.0541$$

This value is needed to find the fugacity of pure water. To do this, we use the CHE Calculator Function, `phiC`, to find the fugacity coefficient of pure water. This is done by inputting the given temperature, pressure, and calculated compressibility while specifying the molar composition range and active component as water. Doing this gets us a fugacity coefficient that we then multiply by the pressure to find the fugacity of pure water to be:

$$f_{H_2O}^v = 0.0184$$

We now need to find the fugacity of the water in the vapor phase of the mixture. To do this, we first find the compressibility of the vapor at the given temperature and pressure. Again, we use the CHE Calculator function, `zcomp`, to find the compressibility; however, this time we specify the molar composition range as z_i instead of just as water. Doing this gets us a compressibility of:

$$z_{mix} = 0.7492$$

This is the compressibility of the mixture, not of water within the mixture. We now need to find the fugacity coefficient of the water within the mixture. We use the function, `phiC`, using the given temperature and pressure, while specifying the molar composition range as z_i and the active component as water. We multiplied this by the pressure and the molar composition of water to find the fugacity of water in the mixture for the vapor phase, $\bar{f}_{H_2O}^v$. This gives the fugacity of water in the mixture; however, this value is for our guessed molar composition of water. We can rewrite the previous formula:

$$f_i^v = \bar{f}_i^v$$

In terms of our calculated fugacity for water:

$$f_{H_2O}^v = \bar{f}_{H_2O}^v$$

To make this equation true, we use Goal Seek to set our calculated fugacity of water in the vapor mixture, $\bar{f}_{H_2O}^v$, equal to the fugacity of the pure water vapor, $f_{H_2O}^v$, by changing the molar composition of the water, z_{H_2O} . Doing this gets us a value of:

$$z_{H_2O} = 0.000765$$

To find the amount of water in lb of water per MMSCF of dry gas (lb/MMSCF), we take the molar composition of water and multiply it by the molecular weight of water in lb per lbmol. We then found the conversion factor to convert from lbmol to MMSCF in the unit conversion section of the CHE Calculator. This conversion factor is:

$$1 \text{ lbmol} = 0.0003795 \text{ MMSCF}$$

Putting all this together gets us:

$$0.000765 \cdot 18.0150 \frac{\text{lb}}{\text{lbmol}} \cdot \frac{1 \text{ lbmol}}{0.0003795 \text{ MMSCF}} = 36.32 \frac{\text{lb}}{\text{MMSCF}}$$

To calculate the Dew Point of the stream, we first need to find the molar composition of the vapor phase of the mixture. We know that for the Dew Point:

$$z_i = y_i$$

We then find the compressibility of the mixture's vapor phase using the CHE Calculators function zcomp. We use y_i for the molar composition range, the given pressure, and a guessed value of the Dew Point Temperature. Determining the compressibility of the mixture enables us to calculate a value for the fugacity coefficient of each component in the vapor phase using the guessed Dew Point Temperature.

To find the Dew Point Temperature, we first need to make an initial guess for the molar composition of the liquid phase. This guess comes from setting the composition of the component with the highest normal boiling point equal to 1 and all other components equal to 0. In the case of our stream, the component with the highest boiling point is water. We then find the compressibility of the liquid phase using zcomp, with x_i for the molar composition range, the given pressure, and a guessed value of the Dew Point Temperature. We then find the fugacity coefficient of each component in the liquid phase using the guessed Dew Point Temperature.

We then use the equation:

$$k_i = \frac{\hat{\phi}_i^L}{\hat{\phi}_i^V}$$

to calculate the k-values of the gas stream. We then set the vapor to feed molar ratio, α , equal to 1. Now that we know the k-values, we were then able to calculate a new molar composition of the liquid phase using the equation:

$$x_i = \frac{y_i}{k_i}$$

After setting all these cells up in the CHE Calculator, we then went to the Dew Point Calculation section in the ribbon and clicked on the Dew Point Temperature button. Clicking this button calculates the Dew Point Temperature that keeps the vapor to feed molar ratio, α , equal to 1, and the Rachford-Rice Equation equal to 0. This only works since

our guessed value for the Dew Point Temperature had the vapor phase compressibility greater than the liquid phase compressibility. After clicking, the Dew Point Temperature is:

$$T_{Dew} = 18.33^{\circ}C \text{ or } T_{Dew} = 65.00^{\circ}F$$

This value is about equal to our given temperature of $65^{\circ}F$

To calculate the Bubble Point of the stream, we first need to find the molar composition of the liquid phase of the mixture. We know that for the Dew Point:

$$z_i = x_i$$

We then find the compressibility of the mixture's liquid phase using the CHE Calculators function `zcomp`. We use x_i for the molar composition range, the given pressure, and a guessed value of the Bubble Point Temperature. Determining the compressibility of the mixture enables us to calculate a value for the fugacity coefficient of each component in the liquid phase using the guessed Bubble Point Temperature.

To find the Bubble Point Temperature, we first need to make an initial guess for the molar composition of the vapor phase. This guess comes from setting the composition of the component with the lowest normal boiling point equal to 1 and all other components equal to 0. In the case of our stream, the component with the lowest boiling point is Methane. We then find the compressibility of the vapor phase using `zcomp`, with y_i for the molar composition range, the given pressure, and a guessed value of the Bubble Point Temperature. We then find the fugacity coefficient of each component in the vapor phase using the guessed Bubble Point Temperature.

We then use the equation:

$$K_i = \frac{\hat{\phi}_i^L}{\hat{\phi}_i^V}$$

to calculate the k-values of the gas stream. We then set the vapor to feed molar ratio, α , equal to 0. Now that we know the k-values, we were then able to calculate a new molar composition of the vapor phase using the equation:

$$y_i = k_i \cdot x_i$$

After setting all these cells up in the CHE Calculator, we then went to the Bubble Point Calculation section in the ribbon and clicked on the Bubble Point Temperature button. Clicking this button calculates the Bubble Point Temperature that keeps the vapor to feed molar ratio, α , equal to 0, and the Rachford-Rice Equation equal to 0. This only works since our guessed value for the Bubble Point Temperature had the vapor phase compressibility greater than the liquid phase compressibility. After clicking, the Bubble Point Temperature is:

$$T_{Bubble} = -57.61^{\circ}C \text{ or } T_{Bubble} = -71.69^{\circ}F$$

Attached below are screenshots of our Excel file used for the calculation of the amount of water in the dry gas stream in lb per MMSCF, as well as the calculations for the Dew and Bubble Point Temperatures.

VLE Calculations			Peng-Robinson							User Calculations		
Fluid Conditions		Mixture	CH4	C2H6	C3H8	C4H10	C4H10	H2O				
Molar Composition		1.0000	0.799357	0.099923	0.039972	0.029977	0.029977	0.000765				
Temperature, C		18.3333										
Pressure, bar		62.0526										
Saturated Vapor Pressure, bar			293.8148	36.5659	7.9985	1.9376	2.9047	0.075				
Bubble Point Pressure, bar												
Dew Point Pressure, bar												
Saturated Vapor Temperature, C												
Bubble Point Temperature, C												
Dew Point Temperature, C												
K-Values												
New K-Values												
Vapor to Feed Molar Ratio (α=V/F)												
The Rachford-Rice Equation												
Vapor Phase		Mixture										
Molar Composition												
New Molar Composition												
Compressibility Factor		0.7492										
Fugacity Coefficient												
Saturated Fugacity Coefficient												
Molar Enthalpy, kJ/kmol												
Molar Entropy, kJ/(kmol K)												
Molar Density, kmol/m3												
Molar Heat Capacity, kJ/(kmol K)												
Liquid Phase		Mixture										
Molar Composition												
New Molar Composition												
Compressibility Factor												
Fugacity Coefficient												
Liquid Activity Coefficient												
Molar Enthalpy, kJ/kmol												
Molar Entropy, kJ/(kmol K)												
Molar Density, kmol/m3												
Molar Heat Capacity, kJ/(kmol K)												
Molar Heat of Vaporization, kJ/kmol												
User Data and Calculations: ON												
Dry Gas wt%		1.0000	0.6095	0.1438	0.0836	0.0827	0.0827					
Dry Gas mol (assume 100 kg)		100.0000	60.9470	14.2560	8.3630	8.2670	8.2670					
Dry Gas kmol		4.7409	3.7827	0.7441	0.3897	0.4422	0.4422					
Dry Gas mol%		1.0000	0.6095	0.1438	0.0836	0.0827	0.0827					
Zcomp Pure Water		0.0541										
Fugacity Coeff Pure Water		0.0003										
Fugacity Pure Water		0.0184										
Fugacity Coeff Water in mix		0.3886										
Fugacity Water in mix		0.0084										

From:

1.0

To:

0.0003795

Unit:

bar

Unit:

mmHg

H2O, lbmol/lbmol dpm

0.0007656

H2O, lbmol/M-HSC dpm

36.3434622

$$x_i = \frac{z_i}{1 + \alpha(K_i^0 - 1)}$$

$$y_i = K_i^0 x_i$$

Figure 1: Problem A, Water in Gas calculations

	A	B	C	D	E	F	G	H	I	J	K	L
1	VLE Calculations		Peng-Robinson									
2	Fluid Conditions	Mixture	CH4	C2H6	C3H8	C4H10	C4H10	H2O				
3	Molar Composition	1.0000	0.7994	0.0999	0.0400	0.0300	0.0300	0.0008				
4	Temperature, C	18.3333										
5	Pressure, bar	62.0526										
6												
7	Saturated Vapor Pressure, bar											
8	Bubble Point Pressure, bar											
9	Dew Point Pressure, bar											
10												
11	Saturated Vapor Temperature, C											
12	Bubble Point Temperature, C	-57.3417										
13	Dew Point Temperature, C											
14												
15	K-Values		1.1816	0.4165	0.1926	0.0909	0.1128	0.0734				
16	New K-Values											
17												
18	Vapor to Feed Molar Ratio ($\alpha=V/F$)	0.0000										
19	The Rachford-Rice Equation	0.0000										
20	Vapor Phase	Mixture										
21	Molar Composition	1.0000	0.9445	0.0416	0.0077	0.0027	0.0034	0.0001				
22	New Molar Composition	1.0000	0.944523	0.041616	0.007698	0.002726	0.003380	0.000056				
23	Compressibility Factor	0.4687										
24	Fugacity Coefficient		0.6827	0.2325	0.0954	0.0398	0.0474	0.0791				
25	Saturated Fugacity Coefficient											
26												
27	Molar Enthalpy, kJ/kmol											
28	Molar Entropy, kJ/(kmol K)											
29	Molar Density, kmol/m3											
30	Molar Heat Capacity, kJ/(kmol K)											
31	Liquid Phase	Mixture										
32	Molar Composition	1.0000	0.799385	0.099923	0.039972	0.029977	0.029977	0.000765				
33	New Molar Composition											
34	Compressibility Factor	0.2117										
35	Fugacity Coefficient		0.8066	0.0968	0.0184	0.0036	0.0053	0.0058				
36	Liquid Activity Coefficient											
37												
38	Molar Enthalpy, kJ/kmol											
39	Molar Entropy, kJ/(kmol K)											
40	Molar Density, kmol/m3											
41	Molar Heat Capacity, kJ/(kmol K)											
42	Molar Heat of Vaporization, kJ/kmol											
43	User Data and Calculations: ON											
44												
45	Bubble Point, F	-71.2150										
46												

	A	B	C	D	E	F	G	H	I	J	K	L
1	VLE Calculations		Peng-Robinson									
2	Fluid Conditions	Mixture	CH4	C2H6	C3H8	C4H10	C4H10	H2O				
3	Molar Composition	1.0000	0.7994	0.0999	0.0400	0.0300	0.0300	0.0008				
4	Temperature, C	18.3333										
5	Pressure, bar	62.0526										
6												
7	Saturated Vapor Pressure, bar											
8	Bubble Point Pressure, bar											
9	Dew Point Pressure, bar											
10												
11	Saturated Vapor Temperature, C											
12	Bubble Point Temperature, C											
13	Dew Point Temperature, C	18.3316										
14												
15	K-Values		26626.32	38931.71	573354.34	9695337.54	28201533.85	0.00				
16	New K-Values											
17												
18	Vapor to Feed Molar Ratio [$\alpha=V/F$]	1.0000										
19	The Rachford-Rice Equation	0.0000										
20	Vapor Phase	Mixture										
21	Molar Composition	1.0000	0.7994	0.0999	0.0400	0.0300	0.0300	0.0008				
22	New Molar Composition											
23	Compressibility Factor	0.7492										
24	Fugacity Coefficient		0.8873	0.5867	0.4166	0.2969	0.3174	0.3886				
25	Saturated Fugacity Coefficient											
26												
27	Molar Enthalpy, kJ/kmol											
28	Molar Entropy, kJ/(kmol K)											
29	Molar Density, kmol/m3											
30	Molar Heat Capacity, kJ/(kmol K)											
31	Liquid Phase	Mixture										
32	Molar Composition	1.0000	0.000030	0.000003	0.000000	0.000000	0.000000	0.999967				
33	New Molar Composition	1.0000	0.000030	0.000003	0.000000	0.000000	0.000000	0.999967				
34	Compressibility Factor	0.0541										
35	Fugacity Coefficient		23625.95	22841.95	238841.63	2878484.33	8950164.18	0.00				
36	Liquid Activity Coefficient											
37												
38	Molar Enthalpy, kJ/kmol											
39	Molar Entropy, kJ/(kmol K)											
40	Molar Density, kmol/m3											
41	Molar Heat Capacity, kJ/(kmol K)											
42	Molar Heat of Vaporization, kJ/kmol											
43	User Data and Calculations: ON											
44												
45	Dew Point, F	64.9969										

Figure 3: Problem A, Dew Point Temperature Calculation

B. In problem B, we are asked to

1. Calculate the Degrees of Freedom by applying the Gibbs Phase Rule.
2. Calculate the solubility of CO₂ in the water phase in mg CO₂ per liter of water.

To calculate the Gibbs Phase Rule D.O.F, we use the formula

$$D.O.F = N + 2 - \pi$$

where

N is the number of components in the system

π is the number of phases in the system.

In our case, we have 5 components and 2 phases, and so the Gibb's Phase Rule D.O.F becomes:

$$D.O.F = 5 + 2 - 2$$

which equals

$$D.O.F = 5$$

To calculate the solubility of CO₂ in the water phase, we must first determine what compounds have the possibility of being dissolved in water. While Argon, Nitrogen, and Oxygen can technically dissolve in water, they will be at such small compositions that we choose to ignore them. We therefore know that in our liquid phase at equilibrium, we will have only CO₂ and O₂. We now have three unknowns, however, which are the composition of water in the vapor phase, the composition of water in the liquid phase, and the composition of carbon dioxide in the liquid phase.

To calculate our unknowns, we start by guessing a composition for water in the total (and therefore the vapor phase as $y_i = z_i$). For this specific problem, we set the initial guess for the composition of water in the vapor as $y_{\text{H}_2\text{O}} = 0.01$. The rest of the compositions in the gas phase change by the factor $y_{\text{new}} = y_{\text{old}} \cdot (1 - y_{\text{H}_2\text{O}})$.

After finding out the molar composition of the vapor phase with water included, we can now move on to finding the compressibility of the vapor phase and the fugacity coefficient of each component. The reason we need the fugacity coefficient is that to find the fugacity of H₂O and CO₂, we must first find the fugacity coefficient. We can then find the fugacity via the formula:

$$f_i = \phi_i \cdot y_i \cdot P$$

where ϕ is the fugacity coefficient, y_i is the molar composition, and P is the overall pressure for the mixture.

Using the above formulas, we find that the fugacity of CO₂ is

$$f_{\text{CO}_2} = 0.0008 \text{ bar}$$

and that the fugacity of H₂O is

$$f_{\text{H}_2\text{O}} = 0.0234 \text{ bar}$$

Now that we have the fugacity of CO₂ and H₂O in the vapor, we can use the fundamental equality of vapor-liquid equilibrium:

$$\bar{f}_i^v = \bar{f}_i^L$$

We also know that the fugacity of CO₂ in the liquid phase is equal to

$$\bar{f}_{\text{CO}_2}^L = H \cdot x_{\text{CO}_2}$$

where H is the Henry's constant of CO₂ in H₂O (we find this using the constants and formula in the book, see attached figure below or Excel sheet) and x_{CO_2} is the molar composition of CO₂ in the liquid, and we know that the fugacity of H₂O in the liquid phase is equal to

$$\bar{f}_{\text{H}_2\text{O}}^L = P^{\text{sat}} \cdot x_{\text{H}_2\text{O}}$$

where P^{sat} is the saturated pressure of water at our temperature of 20 °C, and $x_{\text{H}_2\text{O}}$ is the molar composition of H₂O in the liquid.

Substituting the above formulas into

$$\bar{f}_i^v = \bar{f}_i^L$$

we find that the composition of water and carbon dioxide in the liquid phase at our initial guess ($y_{\text{H}_2\text{O}} = 0.01$) are

$x_i = 5.72\text{E-}07$, and 0.842343761 respectively.

The problem here, however, is that the sum of the two compositions is less than one (0.8423 to be exact), and so we must change our initial guess of $y_{\text{H}_2\text{O}} = 0.01$. To do this, we use the built in feature of Excel, goalseek. We goalseek the mixture summation to be 1 while changing our initial guess. Doing so, we find that

$$y_{\text{H}_2\text{O}} = 0.011872$$

and our final liquid compositions for CO_2 and H_2O are

$$x_i = 5.71\text{E} - 07 \text{ and } 0.999999429 \text{ respectively}$$

Finally, we can calculate the solubility of CO_2 in the liquid water by dividing the molar composition of CO_2 in the liquid, by the molar composition of H_2O in the liquid. Doing this division will return the result of the CO_2 in H_2O . After doing the calculation, we reach that

$$\text{CO}_{2\text{in H}_2\text{O}} = 5.71 \times 10^{-7} \text{ mol}_{\text{CO}_2} / \text{mol}_{\text{H}_2\text{O}}$$

The only problem that remains, however, is that we are in the wrong units. The problem asks for units of mg L^{-1} and we are in $\text{mol}_{\text{CO}_2} / \text{mol}_{\text{H}_2\text{O}}$. To change our units, we multiply by the molar weight of CO_2 and divide by the molar weight of H_2O . This operation results in a value of solubility in $\text{kg}_{\text{CO}_2} / \text{kg}_{\text{H}_2\text{O}}$. To get our final answer, we use the fact that

$$1 \text{ kg} = 1 \text{ L}$$

and we convert our numerator from kg to mg by using

$$1 \text{ kg} = 1\,000\,000 \text{ mg}$$

and finally, we arrive at the answer that the solubility of $\text{CO}_{2\text{in H}_2\text{O}}$ is

$$\boxed{\text{CO}_{2\text{in H}_2\text{O}} = 1.39 \text{ mg L}^{-1}}$$

Attached below are screenshots of the calculations performed in problem B. The first is a screenshot of the VLE calculation sheet and the second is the screenshot of the same sheet, but the user calculation section.

VLE Calculations		Peng-Robinson						User Calculations																																																
Fluid Conditions		Mixture	Ar	N2	O2	CO2	H2O	Values of Henry's constants for some common gases in H ₂ O, as functions of temperature are given below:																																																
1	Molar Composition	1.0000	0.009170	0.771570	0.206973	0.000415	0.011872	<table><thead><tr><th></th><th>A</th><th>B</th><th>C</th><th>D</th></tr></thead><tbody><tr><td>OXYGEN</td><td>144.4080745</td><td>-7775.06</td><td>-18.3974</td><td>-0.00944354</td></tr><tr><td>NITROGEN</td><td>164.9940745</td><td>-8432.77</td><td>-21.558</td><td>-0.00843624</td></tr><tr><td>ARGON</td><td>169.4780745</td><td>-8137.13</td><td>-23.2547</td><td>0.00306357</td></tr><tr><td>HELIUM</td><td>117.3450745</td><td>-4397.06</td><td>-16.2365</td><td>0.00602486</td></tr><tr><td>HYDROGEN</td><td>180.0660745</td><td>-6993.51</td><td>-26.3119</td><td>0.0150431</td></tr><tr><td>CO</td><td>171.7750745</td><td>8296.75</td><td>23.3372</td><td>0</td></tr><tr><td>CO2</td><td>159.8650745</td><td>-8741.55</td><td>-21.669</td><td>0.00110259</td></tr><tr><td>H2S</td><td>149.5490745</td><td>-8226.5</td><td>-20.2307</td><td>-0.00129405</td></tr></tbody></table>					A	B	C	D	OXYGEN	144.4080745	-7775.06	-18.3974	-0.00944354	NITROGEN	164.9940745	-8432.77	-21.558	-0.00843624	ARGON	169.4780745	-8137.13	-23.2547	0.00306357	HELIUM	117.3450745	-4397.06	-16.2365	0.00602486	HYDROGEN	180.0660745	-6993.51	-26.3119	0.0150431	CO	171.7750745	8296.75	23.3372	0	CO2	159.8650745	-8741.55	-21.669	0.00110259	H2S	149.5490745	-8226.5	-20.2307	-0.00129405
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2	Temperature, C	20.0000																																																						
3	Pressure, bar	2.0000																																																						
4	Saturated Vapor Pressure, bar		655.2855	818.1530	674.4648	57.3178	0.0234																																																	
5	Bubble Point Pressure, bar																																																							
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16	New Molar Composition																																																							
17	Compressibility Factor	0.9986																																																						
18	Fugacity Coefficient		0.9980	0.9990	0.9980	0.9921	0.9840																																																	
19	Saturated Fugacity Coefficient																																																							
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24	Liquid Phase	Mixture																																																						
25	Molar Composition	1.0000	0.000000	0.000000	0.000000	5.71E-07	0.99999943																																																	
26	New Molar Composition																																																							
27	Compressibility Factor																																																							
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Figure 4: Problem B, VLE Calculations

42	Molar Heat of vaporization, kJ/kmol						
43	User Data and Calculations: ON						
44	Mol Comps	1.0000	0.009280	0.780840	0.209460	0.000420	
45							
46	A	159.8651					
47	B	-8741.5500					
48	C	-21.6690					
49	D	0.0011					
50	Henry Constant CO2 in Water	1442.5566					
51							
52	Fugacity of CO2 in Vapor	0.0008					
53	Fugacity of H2O in Vapor	0.0234					
54							
55	CO2 in H2O, mg/L	1.39					
56							

Figure 5: Problem B, User Calculations Section

C. This question asks us to

1. Calculate the temperature (in °C) of the inlet and outlet of the turbine
2. Calculate the vapor fraction of the inlet and outlet of the turbine
3. Calculate the power generated by the turbine in kJ kg^{-1} of mixed refrigerant

To begin the solution of this question, we first turn our attention to the inlet side of the turbine. We are told that the refrigerant here is a saturated liquid. Because we are told this, we know that we are at the bubble point of the refrigerant. Knowing this, we can look at the P-T graph provided and find the temperature (guess) at which our pressure will equal 60 bar. Doing so we find an initial bubble point temperature guess of

$$T^o = -95^\circ\text{C}$$

From here, we can follow the algorithm described in the CHE textbook to calculate the true bubble point temperature. This procedure is outlined below.

1. Set the liquid molar composition equal to the overall composition that was given in the question description:

$$z_i = x_i$$

2. Set the lightest (meaning most volatile) vapor molar composition equal to 1, and the rest equal to 0.

$$y_{i \text{ lightest}} = 1$$

3. Calculate the compressibility of the vapor and liquid phases using the CHE Calculator function ZComp
4. Calculate the fugacity coefficient of each vapor and liquid component using the CHE Calculator PhiC (Z in the function is the Zcomp previously calculated)
5. Calculate the K values using the formula

$$K_i = \frac{\phi_i^L}{\phi_i^V}$$

6. set $\alpha = 0$ and solve the Rachford-Rice equation
7. Calculate the new y_i values by the formula

$$y_i = K_i \cdot x_i$$

8. Using the CHE Calculator built in ability to do Bubble Point flash calculations, calculate the Bubble Point Temp by hitting the button in the CHE Calculator, "Bubble Point Temperature"

Using the above algorithm, we were able to determine that the temperature of our inlet was

$$T_{\text{inlet}} = -94.9626^\circ\text{C}$$

We also know that since we are at the bubble point, the vapor fraction is equal to

$$\alpha = 0$$

and we are therefore fully liquid

Using the above algorithm, we also as a side product calculate the molar compositions at the inlet, which we find to be

Table 1: Vapor and Liquid Molar Compositions (Inlet)					
	N ₂	CH ₄	C ₂ H ₆	C ₃ H ₈	C ₄ H ₁₀
Vapor molar composition	0.7064	0.2725	0.0206	0.0000	0.0005
Liquid molar composition	0.1500	0.3000	0.3000	0.0000	0.2500

Tho an important note is that there is no actual vapor in our stream, even though we calculated its molar composition (should a tiny bubble form)

Knowing the vapor fraction, temperature, and pressure, we can now move on to calculating the entropy and enthalpy of the inlet, which will be used to find the total power produced by the turbine later.

To calculate the enthalpy of the inlet, we use the CHE Calculator function Enthalpy (because we were told the refrigerant follows the Peng-Robbinson EOS), doing so, we find that

$$H_{\text{in}} = -98\,285.6010 \text{ kJ kmol}^{-1}$$

and by using the CHE Calculator function Entropy, we find that

$$S_{\text{in}} = -258.5379 \text{ kJ kmol}^{-1} \text{ K}^{-1}$$

We can now move on to the outlet side of the turbine. Here, we are told that the pressure drops to 5 bar as the mixture expands. We must now calculate the temperature and vapor fraction. We guess our initial temperature by again looking at the P-T plot we were provided and we attempt our calculations with a guess of

$$T^o = -120^\circ\text{C}$$

To perform this calculation, we again follow the algorithm outlined in the CHE Calculator textbook and described below:

1. Calculate the saturated vapor pressure for each component using Wilson's formula (in the CHE Calculator, WillsonPsat)
2. Calculate the initial K value guesses by using the formula

$$K_i^o = \frac{P_i^{\text{sat}}}{P}$$

3. Check that setting α to 1 and 0 brackets the root. Then, solve the Rashford-Rice equation for the correct α
4. Calculate the Vapor and Liquid Phase molar compositions using the formulas

$$x_i = \frac{z_i}{1 + \alpha(K_i^0 - 1)} \quad y_i = K_i^0 x_i$$

5. Calculate the compressibility of the vapor and liquid phases using the CHE Calculator function ZComp
6. Calculate the fugacity coefficient of each vapor and liquid component using the CHE Calculator PhiC (Z in the function is the Zcomp previously calculated)
7. Calculate the new K values using the formula

$$K_i^{\text{new}} = \frac{\phi_i^L}{\phi_i^v}$$

8. Click the "Calculate Vapor Fraction" in the P-T Flash Calculations Ribbon to fully iterate and solve for the vapor fraction

After following the above algorithm, we find that at -120 °C, the vapor fraction is equal to:

Table 2: Vapor and Liquid Molar Compositions (Outlet, Temp Guess -120 °C)

	N ₂	CH ₄	C ₂ H ₆	C ₃ H ₈	C ₄ H ₁₀
) Vapor molar composition	0.4455	0.5402	0.0142	0.0000	0.0000
Liquid molar composition	0.0103	0.1865	0.4351	0.0000	0.3681

and we also find that the vapor to feed ratio (the vapor fraction)

$$\alpha = 0.3209$$

We must remember, however, that this is still only for the guessed temperature of -120 °C, and that we must find the true temperature by setting the entropy of the inlet and outlet equal. This is true because we are told that our turbine is perfectly efficient, and so our entropy generated is equal to 0.

However, we must first calculate the entropy and enthalpy at the outlet. To do this, we use the same entropy and enthalpy formulas as before, but we must now perform this calculation for both the liquid and vapor phase as we have both in the stream (because $\alpha = 0.3209$). After calculating the enthalpy and entropy for both the vapor and liquid phase, we can find the total entropy and enthalpy by using the formula:

$$H_{\text{mix}} = \alpha \cdot H^V + (1 - \alpha) \cdot H^L$$

and

$$S_{\text{mix}} = \alpha \cdot S^V + (1 - \alpha) \cdot S^L$$

Doing the above calculations (again, at our guessed temperature of -120 °C), we find that

$$H^{\text{out}} = -98\,379.1158 \text{ kJ kmol}^{-1}$$

and

$$S^{\text{out}} = -254.1203 \text{ kJ kmol}^{-1} \text{ K}^{-1}$$

We find then, that our enthalpy at the inlet and outlet are not equal, and so our guessed outlet temperature is wrong. Iterating through different temperatures (manually as the P-T flash calculations must be performed for each temperature) we find that our final temperature, at which our entropy of the inlet and outlet are equal is

$$T^{\text{outlet}} = -126.625\,984\,21 \text{ °C}$$

Our vapor fraction at this temperature is

$$\alpha = 0.2817$$

our enthalpy and entropy are:

$$H^{\text{out}} = -99\,040.8147 \text{ kJ kmol}^{-1}$$

and

$$S^{\text{out}} = -258.5378 \text{ kJ kmol}^{-1} \text{ K}^{-1}$$

and our vapor and liquid molar compositions are:

Table 3: Vapor and Liquid Molar Compositions (Outlet, Final Temperature)

	N ₂	CH ₄	C ₂ H ₆	C ₃ H ₈	C ₄ H ₁₀
Vapor molar composition	0.4985	0.4936	0.0079	0.0000	0.0000
Liquid molar composition	0.0133	0.2241	0.4146	0.0000	0.3481

We also find that the power output (the difference in enthalpy from the outlet and inlet) is:

$$W = -755.2137 \text{ kJ kmol}^{-1}$$

, which (after we convert using the formula that the $MW_{\text{mix}} = \sum z_i \cdot MW_i$ and multiply $W_{\text{new}} = \frac{W_{\text{old}}}{MW_{\text{mix}}}$), becomes

$$W = -23.1896 \text{ kJ kg}^{-1}$$

Finally, all of the above values have been placed in a table for easy access to below:

Table 4: Summary of Inlet and Outlet Thermodynamic Properties

Property	Inlet	Outlet (Final)
Temperature (°C)	−94.9626	−126.6259
Pressure (bar)	60	5
Vapor fraction α	0	0.2817
Enthalpy (kJ kmol ^{−1})	−98285.6010	−99040.8147
Entropy (kJ kmol ^{−1} K ^{−1})	−258.5379	−258.5378
Power output (kJ kmol ^{−1})	—	−755.2137
Power output (kJ kg ^{−1})	—	−23.1896

Attached below are screenshots of our inlet, and outlet Excel sheets, including the final calculations described above.

	A	B	C	D	E	F	G	H
1	VLE Calculations		Peng-Robinson					
2	Fluid Conditions	Mixture	N2	CH4	C2H6	C3H8	C4H10	
3	Molar Composition	1.0000	0.150000	0.300000	0.300000	0.000000	0.250000	
4	Temperature, C							
5	Pressure, bar	60.0000						
6								
7	Saturated Vapor Pressure, bar							
8	Bubble Point Pressure, bar							
9	Dew Point Pressure, bar							
10								
11	Saturated Vapor Temperature, C							
12	Bubble Point Temperature, C	-94.9626						
13	Dew Point Temperature, C							
14								
15	K-Values		4.7094	0.9084	0.0687	0.0107	0.0019	
16	New K-Values							
17								
18	Vapor to Feed Molar Ratio ($\alpha=V/F$)	0.0000						
19	The Rachford-Rice Equation	0.0000						
20	Vapor Phase	Mixture						
21	Molar Composition	1.0000	0.706408	0.272523	0.020597	0.000000	0.000472	
22	New Molar Composition	1.0000	0.706407	0.272524	0.020597	0.000000	0.000472	
23	Compressibility Factor	0.6584						
24	Fugacity Coefficient		0.8129	0.5401	0.2199	0.1122	0.0535	
25	Saturated Fugacity Coefficient							
26								
27	Molar Enthalpy, kJ/kmol							
28	Molar Entropy, kJ/(kmol K)							
29	Molar Density, kmol/m ³							
30	Molar Heat Capacity, kJ/(kmol K)							
31	Liquid Phase	Mixture						
32	Molar Composition	1.0000	0.150000	0.300000	0.300000	0.000000	0.250000	
33	New Molar Composition							
34	Compressibility Factor	0.2140						
35	Fugacity Coefficient		3.8281	0.4906	0.0151	0.0012	0.0001	
36	Liquid Activity Coefficient							
37								
38	Molar Enthalpy, kJ/kmol	-98285.6010						
39	Molar Entropy, kJ/(kmol K)	-258.5379						
40	Molar Density, kmol/m ³							
41	Molar Heat Capacity, kJ/(kmol K)							
42	Molar Heat of Vaporization, kJ/kmol							

Figure 6: Problem C, Inlet Excel Sheet

VLE Calculations		Peng-Robin					
Fluid Conditions	Mixture	N2	CH4	C2H6	C3H8	C4H10	
Molar Composition	1.0000	0.150000	0.300000	0.300000	0.000000	0.250000	
Temperature, C	-126.6259						
Pressure, bar	5.0000						
Saturated Vapor Pressure, bar		73.6850	9.0205	0.0814	0.0034	0.0002	
Bubble Point Pressure, bar							
Dew Point Pressure, bar							
Saturated Vapor Temperature, C							
Bubble Point Temperature, C							
Dew Point Temperature, C							
K-Values		37.4804	2.2026	0.0191	0.0006	0.0000	
New K-Values		37.4804	2.2026	0.0191	0.0006	0.0000	
Vapor to Feed Molar Ratio ($\alpha=V/F$)	0.2817						
The Rachford-Rice Equation	0.0000						
Vapor Phase	Mixture						
Molar Composition	1.0000	0.498531	0.493559	0.007901	0.000000	0.000008	
New Molar Composition							
Compressibility Factor	0.9411						
Fugacity Coefficient		0.9672	0.9226	0.8258	0.7567	0.6890	
Saturated Fugacity Coefficient							
Molar Enthalpy, kJ/kmol	-42444.2195						
Molar Entropy, kJ/(kmol K)	-70.6011						
Molar Density, kmol/m3							
Molar Heat Capacity, kJ/(kmol K)							
Liquid Phase	Mixture						
Molar Composition	1.0000	0.013301	0.224083	0.414565	0.000000	0.348050	
New Molar Composition							
Compressibility Factor	0.0226						
Fugacity Coefficient		36.2492	2.0322	0.0157	0.0005	0.0000	
Liquid Activity Coefficient							
Molar Enthalpy, kJ/kmol	-121238.79						
Molar Entropy, kJ/(kmol K)	-332.2493						
Molar Density, kmol/m3							
Molar Heat Capacity, kJ/(kmol K)							
Molar Heat of Vaporization, kJ/kmol							
User Data and Calculations: ON							

Figure 7: Problem C, Outlet Excel Sheet

42	Molar Heat of Vaporization, kJ/kmol	
43	User Data and Calculations: ON	
44		
45		
46		
47	Enthalpy Inlet, kJ/kmol	-98285.6010
48	Entropy Inlet, kJ/kmol-K	-258.5379
49		
50	Enthalpy Outlet, kJ/kmol	-99040.8147
51	Entropy Outlet, kJ/kmol-K	-258.5378
52		
53	Entropy Balance	6.92E-05
54		
55	Work, kJ/kmol	-755.2137
56	MW mix, kg/kmol	32.5669
57	Work, kJ/kg	-23.1896
58		

Figure 8: Problem C, Final Value Calculations